

# Population genetics of the endangered Cantabrian capercaillie in northern Spain

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## Keywords

landscape genetics; habitat fragmentation; Bayesian clustering; conservation; *Tetrao urogallus cantabricus*; Cantabrian Mountains.

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Editor: Karen Mock

Received 9 April 2010; accepted 27 October 2010

doi:10.1111/j.1469-1795.2010.00425.x

## Abstract

In this study, we examine the effects of habitat fragmentation on the genetic structure of the Cantabrian capercaillie *Tetrao urogallus cantabricus*, based on eight microsatellite loci in non-invasive samples collected across its entire distribution range in northern Spain. For this purpose, we used Bayesian clustering methods and landscape connectivity analyses. We found low genetic diversity and significant genetic differentiation across the subspecies distribution range. Based on the isolation-by-distance pattern observed, we hypothesized that the low dispersal of the species and the habitat configuration might be shaping the genetic structure of Cantabrian capercaillie, and cause reduced gene flow and diversity of some areas. Three genetic clusters were inferred, one in the northern and two in the southern slope of the Cantabrian Mountains. Our results suggest that small but abundant forest patches on the northern slope of the Cantabrian range do not seem to represent a barrier to gene flow, whereas the more isolated forests in the southern Cantabrian Mountains do represent such a barrier. This result was not concordant with the structure proposed based on lek occupation patterns. Conservation strategies should include restoration of forest connectivity to the extent possible.

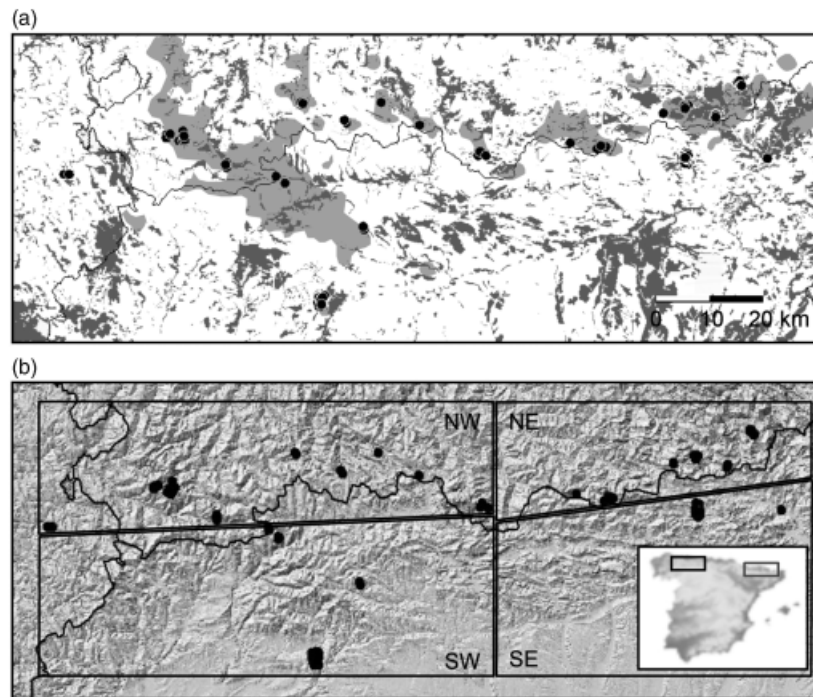
## Introduction

One of the most important issues in conservation biology is fragmentation of one contiguous habitat (Andrén, 1994). Habitat fragmentation can cause population fragmentation, leading to a decreased genetic variability (Frankham, 1996) and an increased likelihood of extinction (Saccheri *et al.*, 1998; Westemeier *et al.*, 1998). However, the importance of habitat connectivity for population persistence is largely unknown for most species (Cushman *et al.*, 2006).

The effect of habitat patches on population genetic structure may depend not only on the straight-line distance between them, or on discrete barriers, but also on the effective cost distance, which reflects the difficulty of individual movement across unsuitable habitat patches in a landscape matrix. This concept of landscape connectivity is used in conservation planning (Taylor, Fahrig & With, 2006), and has recently begun to be applied to the study of the genetic structure of populations, giving rise to the new discipline of 'landscape genetics' (Manel *et al.*, 2003; Coulon *et al.*, 2006; Cushman *et al.*, 2006).

An extreme case of isolation and habitat fragmentation is that of the Cantabrian capercaillie *Tetrao urogallus cantabricus*, Castroviejo (1967) inhabiting the Cantabrian Mountains (NW Spain; Fig. 1). This subspecies is isolated by 300 km from the closest capercaillie population in the

Pyrenees (*Tetrao urogallus aquitanicus* Ingram, 1915). These two taxa represent the southern- and western-most limits of the capercaillie distribution range, and are distinct both ecologically and genetically. For instance, *T. u. cantabricus* is the only subspecies not found in coniferous forest; rather, it lives and feeds almost exclusively in deciduous forests (Obeso & Bañuelos, 2003). Also, the genetic differentiation of Iberian and European capercaillie populations has recently been under investigation in order to determine their taxonomic and conservation status (Duriez *et al.*, 2007; Rodríguez-Muñoz *et al.*, 2007; Segelbacher & Pieltney, 2007). Overall, *Tetrao urogallus* consists of two mitochondrial evolutionary lineages. One lineage is formed by the Iberian populations and the other one by the rest of European populations, which lack genetic structure despite the numerous subspecies described on the basis of morphological and behavioural characteristics (del Hoyo *et al.*, 1994; Duriez *et al.*, 2007). This phylogeographical structure is thought to be derived from an Iberian glacial refugium and a recolonization from eastern refugia, respectively (Segelbacher & Pieltney, 2007). Thus, all the Iberian capercaillie populations constitute a unique Evolutionary Significant Unit and should be managed locally (Duriez *et al.*, 2007). Within the Iberian ESU, *T. u. cantabricus* showed lower genetic diversity than populations in central Europe, while *T. u. aquitanicus* exhibited intermediate diversity



**Fig 1** Maps of the distribution range of the Cantabrian capercaillie *Tetrao urogallus cantabricus*, indicating the samples included in the study. (a) Map of the forest distribution in the Cantabrian Mountains. Forests with a friction value of 1 are shown in dark grey. Lek distribution is shown in light grey (modified from Obeso & Bañuelos, 2003) and black dots represent sampling localities. (b) Map of the Cantabrian Mountain range indicating the sampling sites of *T. urogallus* and their classification into the predefined regions based on lek-use studies. Inset shows the situation of the Iberian capercaillie ESU in Spain: Cantabrian population in black and Pyrenean population in grey.

values and evidence for a putative hybrid or contact zone between the two capercaillie evolutionary lineages (Duriez *et al.*, 2007; Rodríguez-Muñoz *et al.*, 2007).

In Spain, the hunting of capercaillie was prohibited in 1979 but the Pyrenean subspecies was not listed as vulnerable until 2001 (BOE 134, 5 June 2001), and the Cantabrian subspecies as being in danger of extinction until 2005 (BOE 165, 12 July 2005). Currently, the whole population is estimated at 1878–1978 adult individuals, 1378 of which occur in the Spanish Pyrenees (Canut *et al.*, 2003). Furthermore, the conservation strategy for the Cantabrian capercaillie includes the creation of a genetic reserve and a captive breeding programme that was initiated in 2009 (Dirección General para la Biodiversidad, 2004).

The deterioration of the habitat of Cantabrian capercaillie has been indirectly related to its decline (Quevedo, Bañuelos & Obeso, 2006a). The deciduous forests they inhabit have a long history of human use, and today still suffer the pressures of intense logging and cattle grazing (Manuel, Díaz-Fernández & Gil, 2003; Quevedo *et al.*, 2006a). Today, the Cantabrian landscape is highly fragmented due to human impacts (García *et al.*, 2005). The range and numbers of *T. u. cantabricus* populations have been reduced by almost 50% in the last 20 years, with estimates made between 1998 and 2003 as low as 280 and 250 adult males, respectively (Obeso & Bañuelos, 2003; Pollo *et al.*, 2003). Reduction has been particularly pronounced in

smaller habitat fragments, which have been abandoned more readily than large habitat fragments (Quevedo *et al.*, 2006b). This fragmentation of the habitat, and the subsequent abandoning of leks and territories, has been most marked in the eastern and central regions of the Cantabrian Mountains (Obeso & Bañuelos, 2003). Consequently, the capercaillie populations of the southern slope were described as fragmented as early as the 1980s (Pollo *et al.*, 2003). More recently, the dramatic population decline and almost complete absence of leks in the central regions have prompted the idea that the whole Cantabrian capercaillie range could be divided into an eastern and a western core (Obeso & Bañuelos, 2003). Various fragmentation scenarios have been proposed previously for the Cantabrian capercaillie population. An eastern and a western region have been defined based on a population gap at the centre of the range, described in previous habitat and lek studies (Obeso & Bañuelos, 2003; Pollo *et al.*, 2003; Quevedo *et al.*, 2006a). Also, the northern and southern slopes are often considered as different populations, for which independent estimates of abundance are given by the agencies conducting population censuses of capercaillie. Furthermore, the northern slope drops steeply to the Cantabrian Sea coast, while the southern slope descends gradually to the Duero river valley and central Spain's plateau, generating a greater diversity of landscapes and botanical characteristics (Pollo *et al.*, 2003; González *et al.*, 2010).

The capercaillie is generally described as being highly dependent on mature forests and is characterized by low dispersal rates (Gjerde & Wegge, 1989; Swenson & Angelstam, 1993; Storch, 1995; Quevedo *et al.*, 2006b), although other forest types may also be suitable (Rolstad & Wegge, 1990; Rolstad, Rolstad & Wegge, 2007; Miettinen *et al.*, 2009). Therefore, as has been observed for other populations in central Europe (Segelbacher, Manel & Tomiuk, 2008), we expect the Cantabrian capercaillie distribution and genetic structure to be largely conditioned by either the habitat spatial configuration, the geographic distance between individuals or local populations, or both. Therefore, the objective of our study was to characterize the genetic structure across the entire distribution range of the subspecies, and to determine whether there is a significant effect of forest fragmentation and on its genetic diversity.

In order to investigate genetic structuring in the Cantabrian capercaillie and the factors that could be promoting it, we initially used geographically explicit Bayesian clustering methods and compared these results with previously proposed fragmentation scenarios based on habitat and lek use. Secondly, we created a friction map for capercaillie habitat in the Cantabrian Mountains and calculated the Euclidean and effective cost distances between individuals to test which of them better explained the patterns of genetic differentiation.

## Materials and methods

### Sampling and laboratory methods

Samples from the Cantabrian Mountains were collected during summer censuses conducted between 2002 and 2004 across the entire Cantabrian distribution range (Fig. 1a). These samples were mostly composed of moulted feathers, although some faecal samples around leks and samples from predated animals were also obtained.

Total cellular DNA was extracted from either feather quills or a crumbled portion of the non-white end (i.e. urates) of the droppings, using a method based on QIAquick (Qiagen, Crawley, UK) spin columns (Alda, Rey & Doadrio, 2007). Briefly, samples were incubated in 1 mL of extraction buffer (0.5M EDTA pH 8.0, 0.5% SDS, and 50 µg/mL Proteinase K) overnight at 55 °C in a shaking water bath, and the tubes then were centrifuged for 10 min at 2000 × g. After centrifuging, 1 mL of 4.5 M guanidine thiocyanate, 0.5 M potassium acetate pH5.0 buffer was added to the supernatant and mixed. One QIAquick column was set up for each sample. Aliquots (750 µL) of the samples were loaded and the columns were spun for 1 min at 12 800 g. Flowthrough was discarded and the process was repeated until all the sample liquid had passed through the column. Two washes were performed using 700 and 450 µL PE buffer (Qiagen). Finally, the DNA was eluted in 100 µL of TE buffer (10 mM Tris-HCl, pH 8.0, 1 mM EDTA) after a 10 min incubation period at room temperature and 5 min of centrifugation at 12 800 × g.

Capercaillie samples were genotyped for eight microsatellite loci: TUD1, TUD3, TUD6, TUT1, TUT2, TUT3, TUT4 and BG18, using amplification conditions described elsewhere (Segelbacher *et al.*, 2000; Pieltney & Höglund, 2001). Negative controls were included for both the extraction and amplification procedures. PCR products were analysed in the ABI3700 genetic analyser (Applied Biosystems, Foster City, CA, USA). Allele sizes were determined using GeneScan 3.7 and GENOTYPER 3.7 software. To minimize genotyping errors when analysing non-invasive samples arising from their low or degraded DNA contents, faecal samples were genotyped following a multitube approach (i.e. a minimum of three amplifications for each sample, and four additional amplifications if homozygous; Taberlet *et al.*, 1996) and feathers were genotyped at least three times.

### Microsatellite data analysis

Data were checked for null alleles, stuttering and allelic dropout using MICRO-CHECKER (van Oosterhout *et al.*, 2004) and DROPOUT (McKelvey & Schwartz, 2005). Identical genotypes were identified using GIMLET (Valière, 2002). Probabilities of identity for all loci were low enough to allow individual identification ( $PI = 2.508 \times 10^{-9}$  and  $PI_{sib} = 1.26 \times 10^{-4}$ ). In addition, in case of scoring or amplification errors, genotypes differing by only one allele (<5% of all the genotypes obtained) were considered as belonging to the same individual (Regnaut *et al.*, 2006).

### Genetic structure analyses

Current genetic structure across the Cantabrian Mountains was analysed using two complementary Bayesian clustering methods, which do not require that samples be grouped *a priori*, because populations are defined by the genetic data. Firstly, we used the software STRUCTURE 2.1 (Pritchard, Stephens & Donnelly, 2000), which assumes Hardy-Weinberg and linkage equilibrium between loci within populations. We used the admixture model and correlated allele frequencies (F-model; Falush, Stephens & Pritchard, 2003), because we assumed that frequencies in the different populations were likely to be similar. Putative population information was not included in the analysis. We performed 20 runs for each *K*, from *K* = 1 to *K* = 10, with a burn-in of  $5 \times 10^5$  and  $1 \times 10^6$  iterations. Mean log probabilities were used to calculate  $\Delta K$  (i.e. a quantity based on the second-order rate of change of the log probability of data between successive *K* values), and to find the optimal number of *K* following the method of Evanno, Regnaut & Goudet (2005). Once *K* was estimated, the final *Q* coefficients were obtained by averaging the 10 best runs of the selected *K* in CLUMPP 1.1, using the 'greedy' algorithm and the 'all possible input order' options (Jakobsson & Rosenberg, 2007). Secondly, we used GENELAND 2.0.12 (Guillot, Mortier & Estoup, 2005b; Guillot, Santos & Estoup, 2008) in the R-Package (R Development Core Team, 2009). GENELAND uses both the genetic data and geographical information for each individual to estimate population structure. We ran the MCMC

10 times, allowing  $K$  to vary between  $K = 1$  and  $K = 10$ , using the following parameters:  $5 \times 10^5$  iterations, maximum rate of Poisson process fixed at 150, maximum number of nuclei in the Poisson–Voronoi tessellation fixed at 450 and the Dirichlet model for allele frequencies, because its performance is better than that of the F-model (Guillot *et al.*, 2005a). Next, we inferred the number of populations from the modal  $K$  of these runs, and ran MCMC 20 times with  $K$  set at this number and the other parameters unchanged. The posterior probability of population membership was then calculated for each pixel of the spatial domain ( $300 \times 100$  pixels) and the modal population for each individual was calculated for each of the 20 runs (with a burn-in of  $5 \times 10^4$  iterations). The presence of null alleles can affect the accuracy of Bayesian clustering methods, overestimating genetic structure (Guillot *et al.*, 2008). To control for the potential effect of null alleles in our dataset, we repeated the analyses described above using the ‘null allele model’ (Guillot *et al.*, 2008) available in GENELAND 2.0.12.

For each of the STRUCTURE-inferred genetic clusters, we assessed deviations from the Hardy–Weinberg equilibrium by applying exact tests (Guo & Thompson, 1992) in GENEPOP 3.4 (Raymond & Rousset, 1995), following Bonferroni correction (Rice, 1989). The number of alleles ( $N_a$ ), allelic richness ( $A_R$ ), observed and expected heterozygosities ( $H_o$  and  $H_e$ ) and  $F_{IS}$  was calculated using FSTAT 2.9.3 (Goudet, 1995). Also, pairwise  $F_{ST}$  were calculated and an analysis of molecular variance, AMOVA, was performed to test the percentage of genetic variation explained by the inferred genetic clusters. These tests were implemented in ARLEQUIN 3.1 (Excoffier, Laval & Schneider, 2005). Individuals were classified according to the probability of assignment to each cluster inferred by STRUCTURE 2.1.

To test if previously proposed ecological patterns and fragmentation scenarios (see ‘Introduction’; Obeso & Bañuelos, 2003; Pollo *et al.*, 2003; Quevedo *et al.*, 2006a) are consistent with genetic structuring, we classified the specimens by collection site into four regions: NW ( $n = 38$ ), NE ( $n = 33$ ), SW ( $n = 41$ ) and SE ( $n = 39$ ) (Fig. 1b).

Subsequently, we undertook a genetic structure study based on the four predefined Cantabrian regions (NW, NE, SW and SE; Fig. 1b), and tested the existence of an east–west (NE, SE vs. NW, SW) and a north–south (NW, NE vs. SW, SE) population division using an analysis of molecular variance, AMOVA. Recent migration among the four predefined regions was investigated using BAYESASS 1.3 (Wilson & Rannala, 2003). To ensure convergence of the MCMC, we performed 10 runs of  $9 \times 10^6$  iterations including a burn-in of  $3 \times 10^6$  iterations, and a sampling frequency of 2000. Delta values were kept at 0.15 for all parameters, as these rendered acceptance rates within 40–60% of the total.

### Landscape connectivity analyses

To determine if the genetic differentiation observed between Cantabrian capercaillie individuals is better explained through an isolation-by-distance model or by the habitat configuration, we plotted the sampling locations for all

individuals in a geographical information system (GIS) (IDRISI, Eastman, 2003), and obtained the pairwise Euclidean and effective cost distances for all pairs of individuals.

The Euclidean distance was the shortest straight line distance on a map between two individuals, not including elevation. Effective cost distances were estimated with landscape ecological methods included in raster GIS, using the COST PUSH function within the programme IDRISI (Eastman, 2003). To apply these methods, two GIS layers were used: points of interest (capercaillie samples), and a friction map where each pixel had a relative value of ‘movement cost’ for capercaillie, depending on the environmental conditions and landscape elements present in each pixel.

Friction values were set according to known habitat selection patterns for the Cantabrian capercaillie (Quevedo *et al.*, 2006a,b; Bañuelos, Quevedo & Obeso, 2008). Land cover and linear infrastructures (e.g. railways, main roads and local roads) were selected as the most important factors determining the difficulty or cost of movement for capercaillie. Considering that moving 100 m across large patches of forest had a cost of 1 unit, we estimated relative friction values for the different land cover types present in the landscape matrix. Then, we produced a friction map of the studied area by assigning these friction values to the categories of the Corine Landcover 2000 digital map (EEA, 2007), using the second-level legend (i.e. 15 classes; friction values between brackets): forest (1); scrub and/or herbaceous vegetation associations (5); pastures and open spaces with little or no vegetation (15); arable land, permanent crops and heterogeneous agricultural areas (50); local roads and railways (80) and urban fabric, industrial, commercial and transport units, mine, dump and construction sites, water reservoirs and major highways (100). We also tested different friction values, with the same relative importance between classes, and obtained similar results.

Taking into account the altitudinal distribution of capercaillie (Obeso & Bañuelos, 2003) and its tendency to avoid small forest patches (Quevedo *et al.*, 2006b), we introduced two more criteria for assigning friction values: forest patch size and elevation. Forest patches with size  $\leq 10$  ha received a friction value of 5 units (instead of 1), and in the areas with elevation below 700 or above 1700 m we added 10 friction units to the previous values.

Genetic distances, for the complete microsatellite dataset, were calculated as the allele-sharing distance between pairs of individuals (DAS; Nei, 1987), using POPULATIONS 1.2.30 (Langella, 2002).

The presence of an isolation-by-distance or a habitat-related relationship, between the genetic distances and the Euclidean or effective cost distances respectively, was tested using Mantel tests with 10 000 randomizations in ZT 1.1 (Bonnet & Van de Peer, 2002). Also, we tested the effect of possible barriers to gene flow (e.g. ridge, highways, railroads), as they have been proposed to affect the capercaillie distribution along an east–west and a north–south axis (Obeso & Bañuelos, 2003; Pollo *et al.*, 2003; Quevedo *et al.*, 2006a). We performed partial Mantel tests in ZT 1.1 using a binary matrix to code for individuals separated (1), or not

(0), by one of these putative barriers, and the Euclidean distance matrix as a covariate.

## Results

### Genetic diversity inferred from microsatellite loci

Of the 307 samples collected across the Cantabrian Mountains, 212 samples (69%, 197 feathers, 12 faeces and three dead individuals) yielded reliable genotypes based on eight loci and 151 were unique. Seven individuals were captured twice, three were captured four times and six individuals seven or more times. Most of the recaptured genotypes were found in the same location and season. Only in the SE region were the feathers of two males found in 2 consecutive years. All duplicated genotypes were removed from the analyses.

No evidence for linkage disequilibrium was found between loci. MICRO-CHECKER detected signs of allele dropout or null alleles at three loci (TUT1, TUT4 and TUD6), but not consistently among regions or the data overall. DROP-OUT, however, which does not assume HWE, found no evidence of genotyping errors. Overall, the Cantabrian capercaillie's distribution range, the number of alleles per locus ranged from seven for locus TUT3, to 22 for locus TUD6 (average = 11.75). The total number of alleles across loci was 94. Observed heterozygosity per locus ranged from 0.127 to 0.612 with an average value of 0.44 ( $\pm 0.190$ ).

### Bayesian analyses of genetic structure

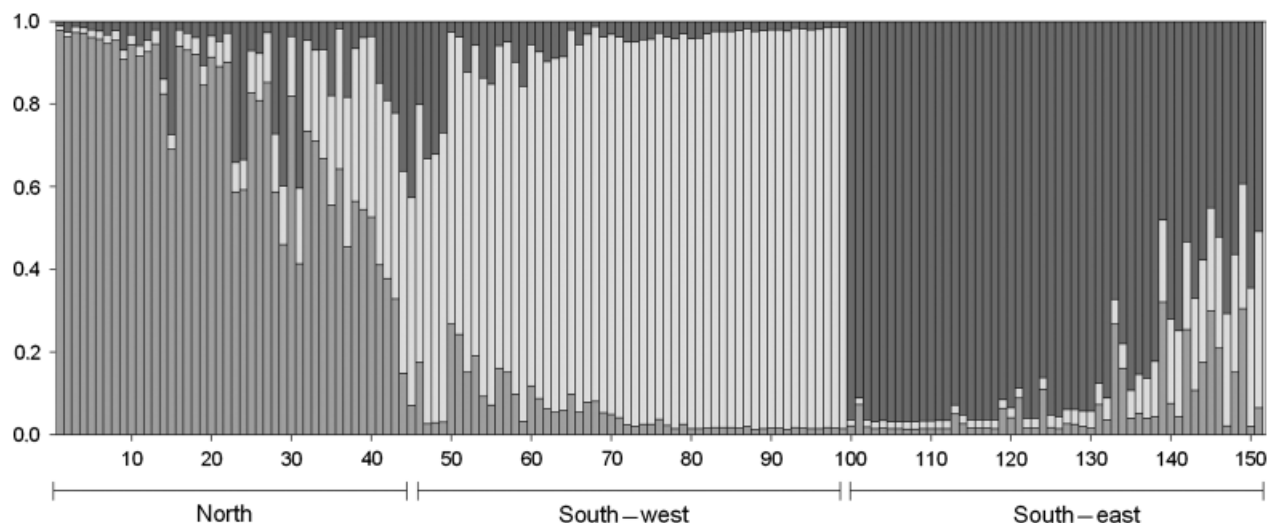
The STRUCTURE analysis allowing admixture and correlated allele frequencies with no prior information about samplings, estimated that the most likely genetic structure for

the Cantabrian capercaillie is  $K = 3$  (S1, S2 and S3) (Fig. 2). Forty-one individuals were assigned to cluster S1, and 52 and 58 individuals were assigned to clusters S2 and S3, respectively (Figs 2 and 3a). The average  $Q$  coefficient for cluster S1 was 0.75, while clusters S2 and S3 showed higher  $Q$  values and consequently less admixture (0.84 and 0.85, respectively) (Fig. 2).

All the runs performed in GENELAND 2.0.12 provided highly congruent results, and all assigned an optimum structure to three clusters (G1, G2 and G3) in over 90% of the MCMC iterations. Similarly, in the analysis performed under the 'null allele model', the existence of three genetic clusters rendered the best structure in 86% of the MCMC iterations, indicating that the null alleles in our dataset, if any, might be in a low proportion ( $< 10\%$ ) and are unlikely to overestimate the genetic structure (Guillot *et al.*, 2008).

We also checked for consistency in the post-processed runs with  $K$  fixed at three. All the posterior probability maps agreed on situating a sharp barrier between the two main groups of samples from the southern slope (i.e. clusters G2 and G3, Fig. 3b). Most of the northern slope constitutes a single large population, cluster G1. However, cluster G2 also includes samples from the north-western part of the distribution range (Fig. 3b). Two individuals genetically assigned to G2 were found in the eastern part of the northern slope, indicating that gene flow is not completely interrupted.

Among the STRUCTURE-defined groups, cluster S1 not only showed the highest values of genetic diversity ( $A_R = 6.980 \pm 2.662$ ,  $H_o = 0.496 \pm 0.135$ ) but also the largest deviations from Hardy–Weinberg equilibrium (Table 1). On the other hand, S2 and S3 showed almost half the number of alleles than S1 ( $N_A = 5.250$ – $5.750$  and  $A_R = 3.442$ – $3.719$ ) and also lower heterozygosity values (Table 1).



**Fig 2** Graphical summary of clustering analysis performed in STRUCTURE for the Cantabrian capercaillie *Tetrao urogallus cantabricus*. Each individual is represented by a vertical line broken into up to three segments representing the estimated proportion of the individual assignment to each of the  $K=3$  clusters. Medium grey corresponds to cluster S1, light grey to cluster S2 and dark grey to cluster S3. The approximate geographic location of each genetic cluster is indicated.



**Fig 3** Maps of the study area indicating the genetic assignment of Cantabrian capercaillie *Tetrao urogallus cantabricus* samples. (a) Geographical representation of the genetic assignment of individuals obtained from *STRUCTURE*. Medium grey circles indicate cluster S1, light grey squares cluster S2 and dark grey triangles cluster S3. Pie diagrams represent the proportion of individuals assigned to each genetic cluster where sampling localities overlap. (b) Map of posterior mode population membership obtained in *GENELAND* for the whole Cantabrian distribution of *Tetrao urogallus*. Medium grey indicates genetic cluster G1, light grey genetic cluster G2 and dark grey represents genetic cluster G3. A question mark indicates that the exact location of the genetic barrier between clusters G2 and G3 is unknown.

AMOVA results showed that the three-cluster arrangement (S1 vs. S2 vs. S3) explained a 4.5% of the genetic variation ( $P < 0.001$ , Table 2). Pairwise  $F_{ST}$  values for the inferred clusters were significant and highest between clusters S1 and S3 ( $F_{ST} = 0.192$ ), and lowest between S1 and S2 ( $F_{ST} = 0.104$ ). The southern clusters (S2 and S3) showed an intermediate value of  $F_{ST} = 0.150$ .

### Genetic structure analyses based on predefined groups

Overall, the four predefined regions (NW vs. NE vs. SW vs. SE) explained a 3.3% of the genetic variation ( $P < 0.001$ ). Low, but significant, genetic differentiation emerged between northern and southern regions ( $F_{CT} = 0.033$ ,  $P < 0.001$ ), but not between the eastern and western ones ( $F_{CT} = 0.110$ ,  $P = 0.171$ , Table 2).

Migration rates, as calculated using *BAYESASS* 1.3, indicated that recent migration between predefined regions mainly occurs from east to west along the northern slope ( $m = 0.194$ ), and from north to south on the western side of the range ( $m = 0.104$ ). In general, the SE region showed the lowest migration rates when compared with the rest of the regions (average immigration rate:  $m = 0.012 \pm 0.008$ ; average emigration rate:  $m = 0.003 \pm 0.000$ , Table 3).

Each of the predefined regions was mainly assigned to one of the inferred clusters (S1, S2, S3 and G1, G2, G3), with the exception of the NW region (Table 4). However, in the southern slope regions, we found higher  $Q$  values for a single cluster than in the northern slope (Table 4). The NW region showed a mixed ancestry of individuals from clusters S1 and S2 (G1 and G2 for the *GENELAND* method). The NE and SW samples were primarily assigned to the S1 and S2 (or G1 and G2) inferred clusters (Table 4).

**Table 1** Genetic diversity of the Cantabrian capercaillie *Tetrao urogallus cantabricus* based on STRUCTURE analysis of microsatellite loci

|            | <i>n</i> | <i>N<sub>a</sub></i> | <i>A<sub>R</sub></i> | <i>H<sub>o</sub></i> | <i>H<sub>e</sub></i> | <i>F<sub>IS</sub></i> |
|------------|----------|----------------------|----------------------|----------------------|----------------------|-----------------------|
| Cluster S1 | 41       | 9.250 (3.382)        | 6.980 (2.662)        | 0.496 (0.135)        | 0.659 (0.169)        | <b>0.267</b>          |
| Cluster S2 | 58       | 5.750 (2.188)        | 3.719 (1.155)        | 0.459 (0.278)        | 0.475 (0.236)        | 0.043                 |
| Cluster S3 | 52       | 5.250 (2.816)        | 3.442 (1.536)        | 0.367 (0.234)        | 0.433 (0.250)        | <b>0.162</b>          |
| ALL        | 151      | 11.750 (4.949)       | 5.048 (1.778)        | 0.440 (0.190)        | 0.568 (0.235)        | <b>0.214</b>          |

*n*, number of samples; *N<sub>a</sub>*, number of alleles; *A<sub>R</sub>*, allelic richness standardized to the minimum sample size; *H<sub>o</sub>*, observed heterozygosity; *H<sub>e</sub>*, expected heterozygosity; *F<sub>IS</sub>*, inbreeding index.

Bold values represent significant deviations from HWE after Bonferroni correction.

**Table 2** Analysis of molecular variance for the Cantabrian capercaillie, based on microsatellite loci

| Partition tested                                                  | % var. among groups | <i>F<sub>IS</sub></i> | <i>F<sub>SC</sub></i> | <i>F<sub>CT</sub></i> |
|-------------------------------------------------------------------|---------------------|-----------------------|-----------------------|-----------------------|
| STRUCTURE K3 (Cluster S1) versus (Cluster S2) versus (Cluster S3) | 4.5                 | <b>0.041</b>          | <b>0.174</b>          | <b>0.045</b>          |
| Four-predefined regions (NW) versus (NE) versus (SW) versus (SE)  | 3.3                 | <b>0.041</b>          | <b>0.167</b>          | <b>0.033</b>          |
| North–south Division [(NW)(NE)] versus [(SW)(SE)]                 | 3.3                 | 0.003                 | <b>0.213</b>          | <b>0.033</b>          |
| East–west division [(NW)(SW)] versus [(NE)(SE)]                   | 11                  | 0.003                 | <b>0.128</b>          | 0.11                  |

*F<sub>IS</sub>*, variation among individuals within localities; *F<sub>SC</sub>*, variation among localities within clusters; *F<sub>CT</sub>*, variation among clusters within the population.

Bold values are significant probabilities  $P < 0.05$ .

**Table 3** Migration rates inferred using BAYESASS among predefined regions in the Cantabrian capercaillie population

|    | NW    | NE    | SW    | SE    |
|----|-------|-------|-------|-------|
| NW | 0.874 | 0.015 | 0.104 | 0.006 |
| NE | 0.194 | 0.771 | 0.014 | 0.021 |
| SW | 0.019 | 0.006 | 0.965 | 0.009 |
| SE | 0.003 | 0.003 | 0.003 | 0.991 |

Values in columns are migration rates into each population from the populations to the left. Values in the diagonal represent proportion of non-migrant individuals.

## Landscape genetic analyses

The results of the Mantel tests indicated a significant relationship between the genetic distance (DAS) and both the Euclidean ( $r = 0.244$ ,  $P = 0.0001$ ) and cost distances ( $r = 0.291$ ,  $P = 0.0001$ ). Overall, the correlation between the DAS distances and the least cost distance matrix was greater than the Euclidean distance (Table 5). However, when different sets of regions were compared, the cost distance did not always explain more variance than Euclidean distance (Table 5).

**Table 4** Mean assignment proportions obtained with the Bayesian clustering methods (STRUCTURE and GENELAND) for the four-predefined regions

|           | Predefined region | Cluster 1<br>S1 | Cluster 2<br>S2 | Cluster 3<br>S3 |
|-----------|-------------------|-----------------|-----------------|-----------------|
| STRUCTURE | NW                | 0.5459          | 0.3280          | 0.1260          |
|           | NE                | 0.8581          | 0.0473          | 0.0946          |
|           | SW                | 0.1726          | 0.7119          | 0.1155          |
|           | SE                | 0.0479          | 0.0481          | 0.9040          |
| GENELAND  | G1                |                 | G2              | G3              |
|           | NW                | 0.4316          | 0.5682          | 0.0002          |
|           | NE                | 0.8668          | 0.0287          | 0.1045          |
|           | SW                | 0.1334          | 0.8140          | 0.0527          |
|           | SE                | 0.0021          | 0.0000          | 0.9979          |

The partial Mantel tests, including the effect of putative barriers for the capercaillie, did not show a significantly better correlation than the Euclidean distances alone (Table 5).

## Discussion

### Genetic structure

The general pattern of genetic structure obtained for the Cantabrian capercaillie population revealed three genetic clusters that roughly partitioned the capercaillie distribution range north and south of the Cantabrian Mountains ridge and subdivided the latter into two groups. A widespread cluster (S1 and G1), covered most of the northern slope; a second cluster (S2 and G2) occupied both slopes in the western Cantabrian range; and a third cluster (S3 and G3) was restricted to the SE region (Fig. 2a and b). The third cluster rendered the highest probabilities for all methods and also the highest  $F_{ST}$  values estimated for the three clusters ( $F_{ST\ S1-S3} = 0.192$ ,  $F_{ST\ S2-S3} = 0.150$ ).

We would expect these genetic divergences to be the result of: (1) low dispersal behaviour; (2), the existence of habitat barriers to gene flow. Some genetic studies of the capercaillie have shown that, at a local scale, there is sufficient gene flow among subpopulations to prevent high genetic differentiation in continuous or fragmented habitats (Segelbacher, Storch & Tomiuk, 2003b; Liukkonen-Anttila *et al.*, 2004), even in small and isolated regions similar to our study area (Regnaut, 2004). On the other hand, in other regions,

**Table 5** Correlation between genetic distance, DAS, and Euclidean and effective cost distances, determined through Mantel tests considering the distribution range as a whole and the genetic clusters inferred with STRUCTURE

| Correlation | Distance                  | Clusters | <i>r</i>     |
|-------------|---------------------------|----------|--------------|
| Euclidean   | Cost                      | All      | <b>0.951</b> |
| DAS         | Euclidean                 | All      | <b>0.244</b> |
|             | Cost                      |          | <b>0.291</b> |
| DAS         | Partial (N–S × Euclidean) | All      | <b>0.244</b> |
|             | Partial (N–S × Cost)      |          | <b>0.291</b> |
| DAS         | Partial (E–W × Euclidean) | All      | <b>0.242</b> |
|             | Partial (E–W × Cost)      |          | <b>0.288</b> |
| DAS         | Euclidean                 | S1       | <b>0.151</b> |
|             | Cost                      |          | <b>0.172</b> |
| DAS         | Euclidean                 | S2 + S3  | <b>0.539</b> |
|             | Cost                      |          | <b>0.532</b> |

The significance of the correlation was determined through 10 000 randomization. Figures in bold indicate significance at  $P \leq 0.05$ .

habitat fragmentation has been associated with significant genetic structure (Segelbacher *et al.*, 2008).

### Low dispersal

The dispersal of the capercaillie is commonly assumed to be low (5–10 km) and female biased (Storch, 1997; Storch & Segelbacher, 2000). However, this short dispersal distance represents a significant percentage of the capercaillie distribution range in the Cantabrian Mountains, where the inhabited area is very small, around 1700 km<sup>2</sup> (Quevedo *et al.*, 2006b). Furthermore, sex-biased dispersal could also condition the results obtained (Storch & Segelbacher, 2000). In this study, we were not able to determine the sex of all the samples; hence, it was not possible to analyse each sex separately. Only the subset of data obtained from male feathers was large enough to be analysed on its own ( $n = 63$ ), and results were congruent with the full dataset including both sexes (data not shown). It is unknown if an exclusively female dataset would show a different result, as females are thought to disperse at higher rates than males (Storch, 1997; Storch & Segelbacher, 2000). However, recent studies based on genetic (Mäki-Petäys *et al.*, 2007) and individual movement data (Carleos & López-Díaz, 2010) contradict this assumption, and suggest that both sexes show similar dispersions and consequently do not differ in their genetic structure.

The significant Mantel correlation between Euclidean and individual genetic distances suggests that isolation by distance could be contributing to the observed genetic structuring. However, the unequal estimated migration rates between predefined regions suggest that these differential migration rates among regions may also contribute to this structuring. In the migration analysis using BAYESASS, the greatest gene flow was estimated to occur from east to west along the northern slope, but was very restricted between the

SW and SE regions. Hence, the unequal rates and the higher immigration towards the NW region could indicate the existence of source-sink dynamics (Segelbacher *et al.*, 2003b), probably due to the abandonment of smaller forest patches at the eastern side of the range and the preferential use of the larger fragments to the west (Quevedo *et al.*, 2006b). Furthermore, the larger differentiation between the southern regions could indicate the existence of a barrier to gene flow, which could be mistaken for an isolation-by-distance effect (Piertney *et al.*, 1998).

### Habitat barriers

Field studies have confirmed previously the existence of discontinuities in the distribution of the Cantabrian capercaillie at the centre of the range (Obeso & Bañuelos, 2003; Pollo *et al.*, 2005). However, none of the previously proposed fragmentation scenarios, on which predefined regions were based, were able to explain a larger proportion of the genetic variation than the inferred genetic clusters (Table 2). Only the effect of the mountain ridge showed a low but significant effect on the differentiation between the northern and southern slope populations (Table 2). Similarly, the partial Mantel tests, which considered the effect of proposed barriers along a north–south or east–west axis failed to improve the correlation with the genetic distance between individuals (Table 5). The same fragmentation pattern has been described in this area for other species of large mammals, but with different genetic effects. The brown bear *Ursus arctos* population is divided into two cores (Nores, 1988), and although only separated by 30 km, there is no evidence of effective dispersal between them and therefore they suffer intense genetic erosion (Pérez *et al.*, 2009). Conversely, the Cantabrian chamois *Rupicapra pyrenaica parva*, despite fragmented populations, shows no genetic differentiation among populations due to its high gene flow (Pérez-Barbería *et al.*, 1996).

On the other hand, the effective cost distance calculated according to the habitat suitability for the capercaillie did show a significant correlation with the genetic distance among individuals (Table 5). This correlation was only slightly better than the one obtained with Euclidean distances, although the observed 5% increase is considered by some authors to be a substantial improvement, considering the large number of potential migration routes between individuals (Vignieri, 2005; Cushman *et al.*, 2006). In any case, these results could support the hypothesis that, besides the main isolation-by-distance effect, there is an underlying effect of the forest dependency of the capercaillie that might also be conditioning dispersal facility along habitat pathways and causing the unequal rates observed (Vignieri, 2005; Segelbacher *et al.*, 2008).

Both the Bayesian methods and those based on predefined regions used to describe genetic structure determined that the severe population decline in the central part of the Cantabrian range (Obeso & Bañuelos, 2003) cannot be detected in terms of genetic subdivision of the population. Only the southern slope seems to be affected by these



discontinuities. However, habitat configuration does not influence genetic structure in the same way for all regions (Table 5). Despite the low habitat suitability of the Cantabrian Mountains as a whole (23% of forest cover), the northern slope has the highest percentage of forest cover (Quevedo *et al.*, 2006a). Consequently, the northern region appears not to present a gene flow barrier for the capercaillie, because it represents a single genetic cluster within which we observed a high migration rate, preferentially in a NE to NW direction. This east–west bias may be due to the better quality of the NW forest fragments because, as it has been proposed, capercaillie might preferentially select the larger forest fragments and abandon or avoid the smaller ones (Obeso & Bañuelos, 2003; Quevedo *et al.*, 2006a).

Conversely, forest cover on the southern slope is much poorer (Quevedo *et al.*, 2006a). While the northern slope sustains a larger number of small forest fragments (over 5000 fragments with an average surface of 14 ha), the southern slope forest cover is divided into fewer, larger fragments further apart from each other (371 fragments with an average surface of over 600 ha) (Obeso & Bañuelos, 2003; García *et al.*, 2005). This forest distribution might represent a disadvantage for the southern populations since, although long-distance dispersals may occur, the greater separation of the southern forests reduces connectivity among fragments. Thus, we assume that longer distances could be covered in the north than in the south, as capercaillie individuals move from one forest patch to another one close-by in a stepping stone fashion. This is supported by the slightly better correlation obtained for the cost distances and the genetic distances than for the Euclidean distances in the northern slope. However, in the southern slope, although the correlation between geographic and genetic distances was much higher than in the north, it was not improved when the habitat configuration was considered, probably due to the total absence of gene flow and the long time of isolation of these regions (Table 5). Furthermore, we should also consider that Cluster 3 (S3 and G3) is the only population of Cantabrian capercaillie occurring in an autochthonous *Pinus sylvestris* forest, and hence possible differential habitat selection for each forest type could condition the dispersal of the capercaillie (Quevedo *et al.*, 2006b).

### Genetic diversity and conservation of the Cantabrian capercaillie

The overall genetic diversity ( $H_e$ ) of the Cantabrian capercaillie was lower than that reported in previous studies, using five to seven microsatellite loci in common, for other northern and central European populations (Segelbacher, Höglund & Storch, 2003a; Mäki-Petäys *et al.*, 2007), and similar to that seen in other endangered or reduced capercaillie populations such as those of the Jura Mountains in Switzerland or the Pyrenees (Regnaut, 2004; Regnaut *et al.*, 2006).

Generally speaking, patterns of low genetic diversity are to be expected as the limit of the distribution range is approached (Ibrahim, Nichols & Hewitt, 1996). In addition,

the reduced genetic diversity of the Cantabrian capercaillie is aggravated by the fact that the population is closed. Therefore, the population fragmentation and decline in numbers would be exaggerated by the effects of drift (Falconer & Mackay, 1996; Frankham, 1996). Accordingly, we detected higher genetic diversity in the northern slope, which besides having the greatest estimated population size (108 occupied leks in the northern slope and 85 leks in the southern; Bañuelos & Quevedo, 2008), was identified as one large genetic cluster. As would be expected, in the smaller populations of the south, diversity was lower (Frankham, 1996), indicating the severe isolation of these regions.

Our results indicate that the Cantabrian capercaillie population is genetically structured at different levels. First, an isolation-by-distance pattern was observed across the whole distributional range and within each region analysed, and that habitat fragmentation also contributed to this pattern. Second, subdivision between the northern and southern populations was observed, likely as a result of the mountain ridge that seems to act as a natural barrier to gene flow. Sylvicultural practices have caused changes in the forest structure of the Cantabrian Mountains that have been negatively associated to the distribution and dynamics of the Cantabrian capercaillie, particularly in the eastern populations (Pollo *et al.*, 2003, 2005), and this has been reflected in the differential migration rates and genetic variability along the distribution range. Also, even though the Cantabrian capercaillie habitat mainly consists of deciduous forests (Obeso & Bañuelos, 2003), other forests types are also occupied. Cluster 3 (S3 and G3) is mainly found in a *P. sylvestris*-dominated forest, and new capercaillie subpopulations have been found in Mediterranean *Quercus pyrenaica* forests of the southern slope (González *et al.*, 2010). Future studies could provide information on possible habitat selection or preferential use of coniferous, Mediterranean or Eurosiberian deciduous forests, and consequently establish management strategies according to habitat characteristics of each region.

Overall, the low genetic differentiation and diversity in the Cantabrian capercaillie population support that this subspecies should be considered as a separate Management Unit within the Iberian capercaillie Evolutionary Significant Unit (Duriez *et al.*, 2007; Segelbacher & Pieltney, 2007) and managed independently (Storch *et al.*, 2006; Duriez *et al.*, 2007; Rodríguez-Muñoz *et al.*, 2007). Most importantly, conservation efforts should be focused to ensure gene flow mainly across the northern slope of the Cantabrian population, to avoid genetic isolation. For this purpose, habitat corridors should be favoured, particularly in the central part of the range and in the lowest areas of the mountain ridge, as these might be used preferentially.

These findings should also be taken into account for the captive breeding programme that collects eggs from the wild to be raised in captivity. For such programmes, the aim is to reinforce the source populations and create a genetic stock for the Cantabrian capercaillie. Therefore, founders should be chosen from different regions to represent the full array of genetic diversity, so that their descendants retain as much

evolutionary potential as possible. However, first and foremost, the factors responsible for the decline of the species should be eliminated. Accordingly, the preservation of the existing mature forest, and particularly the connectivity among them, should be the main concern for any strategy designed to conserve this species.

## Acknowledgements

The authors thank the Consejería de Medio Ambiente y Desarrollo Rural of the Principado de Asturias, and C. Pollo from the Consejería de Medio Ambiente of the Junta de Castilla y León for providing samples and funding. L. Robles, F. Ballesteros and J.L. Benito kindly helped with the sampling. I. Cerón-Souza helped to improve earlier versions of the manuscript, A. Burton reviewed the English text and L. Alcaraz assisted in the laboratory. We also thank the Editor and two anonymous referees for their helpful suggestions. F. Alda benefited from an FPU pre-doctoral grant from the Spanish Ministry of Education and Science.

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